

What is ICT?

Learning Objectives

- Define ICT and identify its components
- List everyday devices that use ICT
- Explain how ICT helps in daily life

Engage (5-7 minutes)

Think of a village grain mandi where farmers bring produce. Instead of counting each grain by hand, machines sort, weigh, and record everything quickly. ICT works like that for information—it sorts, processes, and shares data fast. Today we learn what this powerful tool is and how it touches our lives.

Explore (10-12 minutes)

Look around the classroom and your home. List every device you see that processes information—phones, TVs, calculators, computers. Draw a simple picture of three such devices and write one job each device does. Share your list with a partner.

Explain (10-12 minutes)

ICT stands for Information and Communication Technology. It includes computers, mobile phones, televisions, and the internet—any device that handles information. ICT helps us learn, communicate, and solve problems faster than old methods.

Elaborate (8-10 minutes)

Think about your school. Where does ICT help? The office computer stores student records, the projector shows lessons, and teachers use phones to message parents. Write three ways ICT could help your village more, like sending weather alerts to farmers or tracking bus timings.

Evaluate (5-8 minutes)

Exit ticket: Write one sentence defining ICT and name two devices that are part of it. If you finish early, add one way ICT helps your family.

History of Computers

Learning Objectives

- Identify major generations of computers
- Explain how computers have changed over time
- Compare early computers with modern ones

Engage (5-7 minutes)

Your grandmother might tell you how she used to grind grain by hand, then got a hand mill, and now uses an electric grinder. Computers changed the same way—from room-sized machines to devices in your pocket. Let us trace this journey through time.

Explore (10-12 minutes)

Look at pictures of early computers like ENIAC and modern laptops. Create a simple timeline on paper showing four stages: 1940s room-sized computers, 1970s desktop computers, 1990s laptops, and today's smartphones. Write one feature of each stage.

Explain (10-12 minutes)

Computers evolved through generations. First-generation used vacuum tubes and filled entire rooms. Second-generation used transistors and got smaller. Third-generation used microchips and became personal. Today's computers are powerful, portable, and connected through networks.

Elaborate (8-10 minutes)

Your family might own a smartphone more powerful than early computers that guided rockets. Write a short paragraph comparing what a 1960s computer could do versus your phone today. Discuss with a partner how this change affects your studies.

Evaluate (5-8 minutes)

Exit ticket: Name two ways modern computers are different from early ones. Draw a quick sketch of the oldest and newest computer you can think of.

Introduction to Internet

Learning Objectives

- Define the Internet and its basic function
- Explain how devices connect through networks
- Identify services available on the Internet

Engage (5-7 minutes)

Imagine a bus route connecting your village to the town, the district headquarters, and the state capital. The Internet works like a vast network of roads connecting computers worldwide. Today we explore how this digital highway works and what travels on it.

Explore (10-12 minutes)

Draw a simple map with your school, a friend's house in another village, and a city. Draw lines connecting them like roads. Label these lines as "Internet connections." Write what information could travel on these lines—messages, photos, news.

Explain (10-12 minutes)

The Internet is a global network connecting millions of computers. It uses cables, wireless signals, and satellites to send data. Services like email, websites, video calls, and file sharing all travel through this network.

Elaborate (8-10 minutes)

Think about video calls with relatives in cities or watching educational videos on YouTube. These all use the Internet. Write three Internet services you use and explain how each helps you. Discuss with your group which service is most important for your village.

Evaluate (5-8 minutes)

Exit ticket: Define the Internet in one sentence and list two services it provides. If you use the Internet at home, name one thing you do on it.

Web Browsers

Learning Objectives

- Identify popular web browsers and their icons
- Open a browser and navigate to a website
- Explain the parts of a browser window

Engage (5-7 minutes)

When you visit the gram panchayat office, different clerks handle different tasks—one records births, another manages land records. Web browsers are like those clerks for the Internet, each handling how you view websites. Today we meet these helpful tools.

Explore (10-12 minutes)

Look at the computer screens. Identify the browser icons—Chrome, Firefox, Edge. Count how many different browsers you can find. Open one browser and explore its parts: address bar, back button, refresh button. Draw a labeled diagram of the browser window.

Explain (10-12 minutes)

A web browser is software that lets you access websites. Popular browsers include Google Chrome, Mozilla Firefox, and Microsoft Edge. The address bar is where you type website addresses, and navigation buttons help you move between pages.

Elaborate (8-10 minutes)

Try opening two different browsers and visiting the same website. Notice any differences in how pages look. Write about your observation. Discuss with your partner which browser feels easier to use and why.

Evaluate (5-8 minutes)

Exit ticket: Name two web browsers and identify one part of a browser window. Show how to refresh a page on the computer.

Search Engines

Learning Objectives

- Define search engines and name popular ones
- Perform a basic search with keywords
- Evaluate search results for relevance

Engage (5-7 minutes)

When you need specific information from the village library, you tell the librarian what you are looking for, and she finds the right book. Search engines work like wise librarians for the Internet—they find information when you ask the right questions. Let us learn how to ask them properly.

Explore (10-12 minutes)

Open a browser and go to Google. Search for “weather in Bangalore.” Write down the first three results you see. Now search for “weather in your village name.” Compare the results. Which search gave better information? Why?

Explain (10-12 minutes)

Search engines like Google, Bing, and Yahoo help you find information on the Internet. You type keywords related to what you need, and the engine shows relevant websites. Using specific words gives better results than vague ones.

Elaborate (8-10 minutes)

Practice searching for different topics—famous temples in Karnataka, farming seasons, or railway timings. Try using two or three specific words each time. Write a tip list for good searching: what works and what does not.

Evaluate (5-8 minutes)

Exit ticket: Name one search engine and demonstrate searching for “Dussehra festival.” Write one tip for finding good information online.

URLs and Websites

Learning Objectives

- Define URL and identify its parts
- Distinguish between websites and web pages
- Navigate using URLs in a browser

Engage (5-7 minutes)

Every house in your village has a unique address—house number, street, village, and pin code. Websites have similar addresses called URLs. Today we learn to read these digital addresses and find our way around the web.

Explore (10-12 minutes)

Look at these URLs: www.google.com, www.wikipedia.org, www.karnataka.gov.in. Write them down and identify the parts after the dot—.com, .org, .gov. What do these endings tell you about the website? Match each URL to what kind of site it might be.

Explain (10-12 minutes)

URL stands for Uniform Resource Locator—it is the address of a website. It has parts like protocol (http/https), domain name, and extension (.com, .org, .gov). Websites contain many web pages, and each page has its own specific URL.

Elaborate (8-10 minutes)

Visit a government website like karnataka.gov.in and a learning site like wikipedia.org. Compare their URLs and content. Write about the differences you notice. Discuss which types of websites are most useful for students in your village.

Evaluate (5-8 minutes)

Exit ticket: Write the full form of URL and identify the domain extension of www.google.com. Name one government website you visited.

Introduction to GPS

Learning Objectives

- Define GPS and explain how it works
- Identify devices that use GPS technology
- Describe real-life applications of GPS

Engage (5-7 minutes)

When a bus driver travels from your village to the city, he follows road signs and landmarks to know where he is. GPS is like invisible road signs in the sky that tell devices exactly where they are. Let us explore this amazing technology.

Explore (10-12 minutes)

Look at a physical map of your district. Mark your village, the nearest town, and the district headquarters. Now think about how a phone knows which of these three places you are at. Write your ideas on paper and share with the class.

Explain (10-12 minutes)

GPS stands for Global Positioning System. It uses satellites orbiting Earth to send signals to devices like phones and car navigation systems. By receiving signals from multiple satellites, a device calculates its exact location—latitude, longitude, and altitude.

Elaborate (8-10 minutes)

GPS helps in many ways: auto-rickshaws find addresses, delivery trucks track routes, farmers use precision farming, and ambulances reach patients faster. Write three ways GPS could help people in your village specifically, like finding lost cattle or guiding emergency vehicles.

Evaluate (5-8 minutes)

Exit ticket: Write the full form of GPS and explain in one sentence how it works. Name one device that uses GPS.

Google Earth

Learning Objectives

- Open Google Earth and navigate the interface
- Search for locations and explore them virtually
- Measure distances between two points

Engage (5-7 minutes)

Imagine you could fly like a bird and see your village from the sky, then zoom out to see all of Karnataka, India, and finally the whole Earth. Google Earth lets you do exactly that—explore our planet from your computer screen. Let us take this virtual flight today.

Explore (10-12 minutes)

Open Google Earth in the browser. Search for your school's location. Zoom in slowly until you can see the building. Now zoom out to see your village, then your district. Write down at what zoom level you can first see your village name. Explore three neighboring towns.

Explain (10-12 minutes)

Google Earth is a virtual globe program that shows satellite images of Earth. You can zoom from space to street level, search for any place, and even see 3D buildings. It uses satellite imagery, aerial photography, and GIS data to create this detailed view.

Elaborate (8-10 minutes)

Try measuring the distance from your school to the nearest railway station using the ruler tool. Compare it with the road distance you know. Explore famous landmarks like the India Gate or Mysore Palace. Write about what you discovered.

Evaluate (5-8 minutes)

Exit ticket: Demonstrate opening Google Earth and searching for “Bangalore.” Use the ruler tool to measure distance between two points. Write one thing you learned about your area.

Internet Safety

Learning Objectives

- Identify common online dangers
- Practice safe browsing habits
- Create a strong password

Engage (5-7 minutes)

When you walk to school, you avoid strangers, look both ways before crossing the road, and stay with friends. The Internet needs similar caution—there are digital dangers too. Today we learn how to stay safe while exploring the online world.

Explore (10-12 minutes)

Read these scenarios: someone asks for your photo online, a website asks for your bank details, you get a message saying you won a prize. Discuss with your partner what you should do in each case. Write your safety rules on paper.

Explain (10-12 minutes)

Internet safety means protecting yourself from online dangers like strangers asking personal questions, fake websites stealing information, and harmful content. Never share personal details like address, phone number, or photos with strangers online.

Elaborate (8-10 minutes)

Practice creating strong passwords. Try combinations like your village name plus birth year plus symbols. Discuss with your group which passwords are easy or hard to guess. Create a poster of five Internet safety rules for your classroom.

Evaluate (5-8 minutes)

Exit ticket: Write three rules for staying safe online. Create one strong password using letters, numbers, and symbols. Explain why we should not share personal information with strangers.

Email Basics

Learning Objectives

- Identify the parts of an email address
- Compose and send a simple email
- Explain the purpose of inbox, sent, and draft folders

Engage (5-7 minutes)

Sending a letter through the post office takes days-writing the address, putting a stamp, and waiting. Email is like instant post-you write a message and it reaches the recipient in seconds. Let us learn how this digital post system works.

Explore (10-12 minutes)

Look at this email address: ramesh.kumar@gmail.com. Write it down and identify the parts: username (before @), the @ symbol, and the domain (after @). Now write three different email addresses using your own names and different domains like yahoo.com or rediffmail.com.

Explain (10-12 minutes)

Email stands for Electronic Mail. An email address has three parts: username, @ symbol, and domain. When you send an email, it goes to the recipient's inbox on their email server. You can also save drafts and check sent messages.

Elaborate (8-10 minutes)

Open a email service and create a simple email. Write a message to your teacher about what you learned today. Practice using CC to send a copy to a friend. Explore the inbox, sent, and draft folders to understand where emails are stored.

Evaluate (5-8 minutes)

Exit ticket: Write the full form of email and label the three parts of ramesh@gmail.com. Explain what the inbox and sent folders do.

Downloading Files

Learning Objectives

- Download files from the Internet safely
- Identify file types and their extensions
- Organize downloaded files in folders

Engage (5-7 minutes)

When you collect seeds from the market, you bring them home and store them properly in labeled containers—tomato seeds here, chili seeds there. Downloading files is similar—you bring digital items from the Internet and organize them on your computer. Let us learn how.

Explore (10-12 minutes)

Search for “Karnataka state map” on Google Images. Right-click on a map and choose “Save image as.” Note where it saves by default. Now search for a PDF worksheet on any topic. Download it and note the file type. Compare how images and documents save differently.

Explain (10-12 minutes)

Downloading means copying a file from the Internet to your computer. Files have extensions like .jpg for images, .pdf for documents, and .mp4 for videos. Always download from trusted websites and scan files for viruses before opening them.

Elaborate (8-10 minutes)

Create a folder called “My Downloads” on the desktop. Download three different files—an image, a document, and a music file. Move them into your folder and rename them appropriately. Discuss with your partner how organized files are easier to find.

Evaluate (5-8 minutes)

Exit ticket: Write the steps to download an image from the Internet. Name three file extensions and their file types. Show how to create a folder and move a file into it.

Chapter 1 Review

Learning Objectives

- Review key concepts from Chapter 1
- Demonstrate skills learned in previous lessons
- Identify areas needing more practice

Engage (5-7 minutes)

Before the harvest festival, farmers check their tools, review their work, and prepare for the next season. Today we do the same with what we learned about the Internet and ICT. Let us review, practice, and celebrate our knowledge.

Explore (10-12 minutes)

Work in groups of four. Each group gets a review card with questions about ICT, Internet, browsers, search engines, URLs, GPS, Google Earth, safety, email, and downloading. Discuss answers and prepare to present one topic to the class. Help each other with difficult questions.

Explain (10-12 minutes)

Chapter 1 covered the basics of ICT and Internet. We learned about devices, history, browsers, search engines, URLs, GPS, Google Earth, safety, email, and downloading. Each concept connects to help us use technology responsibly and effectively.

Elaborate (8-10 minutes)

Create a mind map showing all Chapter 1 topics and how they connect. Start with “ICT” in the center and branch out to each lesson topic. Write one real-life application for each topic. Present your mind map to the class and explain the connections.

Evaluate (5-8 minutes)

Exit ticket: Choose three topics from Chapter 1 and write one key fact about each. Identify which topic you found most interesting and which needs more practice. Rate your confidence level from 1 to 5.

Introduction to Graphics

Learning Objectives

- Define graphics and identify their types
- Distinguish between raster and vector graphics
- Name software used for creating graphics

Engage (5-7 minutes)

When a farmer paints his house for Dasara, he uses a brush with fixed patterns. But a artist drawing Rangoli can create any design from scratch. Graphics work similarly on computers—some are made of tiny dots, others of mathematical shapes. Let us explore both types.

Explore (10-12 minutes)

Look at two pictures on the screen: a photograph of a flower and a drawing of the same flower made with shapes. Zoom in on both until you see pixels on the photograph but the drawing stays smooth. Draw these observations in your notebook and label which is which.

Explain (10-12 minutes)

Graphics are visual images created on computers. Raster graphics use tiny colored dots called pixels—like photographs. Vector graphics use mathematical formulas to create shapes—like logos and drawings. Each type has different uses and qualities.

Elaborate (8-10 minutes)

Think about where you see graphics daily: textbook illustrations, movie posters, festival banners, shop logos. Categorize five examples as either raster or vector. Discuss with your partner why a shop might choose one type over another for their signboard.

Evaluate (5-8 minutes)

Exit ticket: Write the difference between raster and vector graphics in one sentence each. Name one software for creating graphics and one real-life example of each type.

Paint Software Basics

Learning Objectives

- Open Paint and identify its interface components
- Use basic tools like pencil, brush, and eraser
- Save a drawing in Paint

Engage (5-7 minutes)

Think of a blank slate used by village potters to shape clay into pots and vessels. MS Paint is like that blank slate for digital art—it gives you tools and a canvas to create anything you imagine. Let us learn to use this creative playground.

Explore (10-12 minutes)

Open MS Paint on the computer. Look around the interface and identify the ribbon at the top with tools, the white canvas in the center, and the color palette. Click on each tool—pencil, brush, eraser—and see what happens when you draw on the canvas. Write down what each tool does.

Explain (10-12 minutes)

MS Paint is a simple drawing program that comes with Windows. The ribbon contains all tools organized in groups. The canvas is your drawing area. The color palette lets you choose colors. You can resize the canvas, undo mistakes, and save your work.

Elaborate (8-10 minutes)

Draw a simple village scene using only the pencil tool. Now redraw it using brushes for different effects. Add colors from the palette. Save your work as “my_village.jpg” on the desktop. Compare your drawing with a partner and discuss which tools worked best.

Evaluate (5-8 minutes)

Exit ticket: Name three tools in MS Paint and what each does. Show how to save a drawing. Write one advantage of using Paint for beginners.

Drawing Tools

Learning Objectives

- Use shape tools to create geometric figures
- Apply line tools with different thicknesses
- Combine shapes to create simple compositions

Engage (5-7 minutes)

When village carpenters build furniture, they use different saws and chisels for different cuts—straight, curved, detailed. Drawing tools in Paint work the same way—each tool creates a specific shape or line. Let us master these digital tools.

Explore (10-12 minutes)

Open Paint and explore the shapes section. Draw a rectangle, circle, triangle, and star. Change line thickness using the Size option. Draw the same shapes with thin and thick lines. Now combine shapes to draw a house with a rectangular body and triangular roof.

Explain (10-12 minutes)

Paint offers shape tools for rectangles, circles, triangles, and stars. Line tools create straight lines with adjustable thickness. You can fill shapes with colors and draw freehand with pencil and brush tools. Combining these basics creates complex drawings.

Elaborate (8-10 minutes)

Create a simple landscape using only shapes: rectangle for buildings, circles for sun and moon, triangles for mountains, lines for roads. Add colors to make it vibrant. Try the fill tool to color large areas quickly. Share your creation with the class.

Evaluate (5-8 minutes)

Exit ticket: Draw and label three shapes using Paint's shape tools. Show how to change line thickness. Create a simple composition using at least two different shapes.

Color Selection

Learning Objectives

- Identify primary and secondary colors
- Use the color palette and color picker
- Apply colors to shapes and backgrounds

Engage (5-7 minutes)

Think of mixing haldi and kumkum to create new colors for festival decorations, or blending natural dyes for cloth. Digital color selection works similarly—primary colors mix to create secondary ones. Let us explore the color world in Paint.

Explore (10-12 minutes)

Open the color palette in Paint. Identify the basic colors: red, blue, yellow, green. Click “Edit colors” to see the color mixer. Mix red and yellow to make orange, blue and yellow to make green. Draw shapes and fill each with a different color. Count how many colors you can create.

Explain (10-12 minutes)

Primary colors (red, blue, yellow) mix to create secondary colors (green, orange, purple). The color palette shows basic colors, and the color picker lets you create custom shades by adjusting RGB values. You can apply colors to both shapes and backgrounds.

Elaborate (8-10 minutes)

Draw a rainbow using arcs and color each band with the correct color. Create a simple Indian flag with saffron, white, and green. Try creating your school’s colors using the color mixer. Discuss with your partner why understanding colors matters for designers.

Evaluate (5-8 minutes)

Exit ticket: Name three primary colors and three secondary colors. Show how to use the color picker to create a custom color. Draw and color a simple shape using two different colors.

Symmetric Designs

Learning Objectives

- Create mirror images using the select tool
- Design patterns with rotational symmetry
- Apply symmetry to real-world design problems

Engage (5-7 minutes)

Look at Rangoli designs on your doorstep—they are perfectly balanced, with one side mirroring the other. Symmetry makes designs beautiful and balanced. Today we learn to create these mirror patterns digitally in Paint.

Explore (10-12 minutes)

Draw a simple butterfly shape on one side of the canvas. Use the select tool to select it, copy it, and paste it. Flip the pasted copy horizontally and move it to the other side. Observe how the two halves create a symmetric design. Try this with a leaf shape too.

Explain (10-12 minutes)

Symmetry means both sides of a design are mirror images. In Paint, you can create symmetry by drawing one half, copying it, flipping it, and placing it opposite. Rotational symmetry means the design looks the same when rotated. These principles are used in art, architecture, and nature.

Elaborate (8-10 minutes)

Create a symmetric Rangoli pattern using shapes and colors. Start with a central circle, add identical petals around it, and color them symmetrically. Try creating a symmetric house design or a face. Compare your designs with classmates and appreciate the symmetry.

Evaluate (5-8 minutes)

Exit ticket: Explain what symmetry means in your own words. Show the steps to create a mirror image in Paint. Draw a simple symmetric design using at least three shapes.

Image Editing

Learning Objectives

- Crop images to remove unwanted areas
- Resize images while maintaining proportions
- Adjust brightness and contrast of images

Engage (5-7 minutes)

When you harvest vegetables, you trim the leaves and unwanted parts to keep only the good ones. Image editing is similar—you remove unwanted parts and adjust quality to make pictures better. Let us learn these essential editing skills.

Explore (10-12 minutes)

Find an image file on the computer and open it in Paint. Use the select tool to draw a box around the part you want to keep. Click “Crop” to remove everything outside. Now resize the image by dragging corners. Observe how the proportions change. Try making it smaller and larger.

Explain (10-12 minutes)

Cropping removes unwanted parts of an image, keeping only what you need. Resizing changes image dimensions—always maintain proportions to avoid distortion. Brightness controls how light or dark an image appears, while contrast adjusts the difference between light and dark areas.

Elaborate (8-10 minutes)

Take a group photograph and crop it to show only two friends. Resize it to fit a passport photo size. Adjust brightness if the photo is too dark. Create before-and-after versions and save both. Discuss with your partner which edits are most commonly needed.

Evaluate (5-8 minutes)

Exit ticket: Show how to crop an image to keep only the center portion. Explain why we maintain proportions when resizing. Write one situation where adjusting brightness would help.

Layers Concept

Learning Objectives

- Understand what layers are in graphics software
- Create and manage multiple layers
- Arrange layer order to control visibility

Engage (5-7 minutes)

Think of a traditional puppet show—the background scenery is painted first, then characters are placed in front. Layers work the same way in digital art—each element sits on its own transparent sheet. Let us learn to build pictures layer by layer.

Explore (10-12 minutes)

Open a graphics program with layer support. Create three layers: background, middle ground, and foreground. Draw a sky on the background layer, hills on the middle, and a tree on the front. Observe how each layer is independent. Try hiding one layer and see what happens.

Explain (10-12 minutes)

Layers are like transparent sheets stacked on top of each other. Each layer holds separate parts of a drawing—background, characters, text. You can edit, hide, or rearrange layers without affecting others. This makes complex designs manageable and editable.

Elaborate (8-10 minutes)

Create a simple landscape with at least four layers: sky, mountains, river, and foreground elements. Add a sun on a separate layer and experiment with moving it behind mountains. Discuss with your group why layers make designing easier than drawing everything on one layer.

Evaluate (5-8 minutes)

Exit ticket: Explain what a layer is using the transparent sheet analogy. Show how to create, hide, and reorder layers. Draw a simple scene using at least three separate layers.

Filters and Effects

Learning Objectives

- Apply basic filters like blur and sharpen
- Use artistic effects to transform images
- Compare original and filtered versions

Engage (5-7 minutes)

When you look through colored glass or a camera with special lens, the world looks different—warmer, cooler, softer. Filters in graphics software work the same way—they change how images appear with simple clicks. Let us explore these visual transformations.

Explore (10-12 minutes)

Open an image in a photo editor. Apply different filters one by one: blur, sharpen, sepia, grayscale. Take note of how each changes the image. Save the original, then create three versions with different effects. Compare all four and write which effect you like best.

Explain (10-12 minutes)

Filters and effects are preset adjustments that change image appearance. Blur softens details, sharpen enhances edges, sepia gives a brownish vintage look, and grayscale removes color. These effects help create different moods and artistic styles in images.

Elaborate (8-10 minutes)

Take a portrait photograph and apply five different effects. Create a gallery of versions showing the same image in different styles. Discuss with your partner which effects would work best for a school magazine cover versus a science project illustration.

Evaluate (5-8 minutes)

Exit ticket: Apply three different effects to an image and label each. Explain what blur and sepia effects do. Write one situation where you would use a grayscale effect.

Text in Graphics

Learning Objectives

- Add text to graphics using text tools
- Format text with fonts, sizes, and colors
- Create simple banners and posters with text

Engage (5-7 minutes)

Look at festival banners in your village—they combine words and pictures to send messages. Adding text to graphics makes communication stronger. Today we learn to blend words with images to create meaningful designs.

Explore (10-12 minutes)

Create a new canvas in Paint. Click the Text tool (A icon) and click on the canvas. Type your name. Now experiment with different fonts, sizes, and colors from the text toolbar. Create three versions of your name in different styles. Which looks best for a school project?

Explain (10-12 minutes)

The text tool lets you add words to graphics. You can choose fonts (different letter styles), sizes (how big or small), colors, and effects like bold or italic. Text can be placed anywhere on the canvas and edited later. Good text placement makes designs readable and attractive.

Elaborate (8-10 minutes)

Design a simple poster announcing your school annual day. Include the event name, date, and venue using different font styles. Make the title large and bold, details smaller. Add a decorative border. Show your poster to classmates and get feedback on readability.

Evaluate (5-8 minutes)

Exit ticket: Show how to add text to a graphic and change its font and color. Create a simple banner with a title and one line of information. Explain why font choice matters for readability.

File Formats

Learning Objectives

- Identify common image file formats
- Compare qualities and uses of different formats
- Save files in appropriate formats for specific purposes

Engage (5-7 minutes)

When you buy rice, you choose between different packaging—gunny bags for bulk, plastic packets for retail, airtight containers for storage. Image files also need different “packaging” depending on their use. Let us learn about these digital containers.

Explore (10-12 minutes)

Create the same drawing in Paint and save it in three formats: BMP, PNG, and JPEG. Check the file sizes—notice how they differ. Now resize each to 50% and save again. Compare quality and size differences. Write your observations about which format is smallest and which is best quality.

Explain (10-12 minutes)

Common image formats serve different purposes. BMP stores every pixel but creates large files. PNG keeps quality with smaller sizes and supports transparency. JPEG compresses well for photos but loses some quality. GIF works for simple animations. Choosing the right format depends on your needs.

Elaborate (8-10 minutes)

Think about when to use each format: a school project needs high quality (PNG), a website needs small files (JPEG), a logo needs transparency (PNG), a fun animation (GIF). Create a chart matching formats to uses. Discuss why format choice matters for sharing online.

Evaluate (5-8 minutes)

Exit ticket: Name three image formats and write one use for each. Explain why JPEG files are smaller than BMP. Show how to save a file in PNG format.

Project - School Logo

Learning Objectives

- Apply all learned graphics skills to create a logo
- Plan and execute a design project step by step
- Present and explain design choices to others

Engage (5-7 minutes)

Every school has a symbol that represents its values and identity. Creating a school logo is like putting your school's spirit into a single image. Today we combine everything we learned to create something meaningful for our school.

Explore (10-12 minutes)

First, brainstorm ideas with your group. What represents your school? A book, a lamp, a tree, your building? Sketch three rough ideas on paper. Choose the best one and plan which tools, colors, and shapes you will need. Write your design plan before touching the computer.

Explain (10-12 minutes)

A good logo is simple, memorable, and meaningful. It uses shapes, colors, and text effectively. Planning before designing saves time and creates better results. Professional designers always sketch ideas first, then refine them digitally.

Elaborate (8-10 minutes)

Create your school logo using Paint or a graphics program. Use shapes for the main design, add your school name with text tools, apply appropriate colors, and ensure it looks good at different sizes. Save it in PNG format. Prepare to explain your design choices to the class.

Evaluate (5-8 minutes)

Exit ticket: Present your logo and explain three design choices: why you chose those shapes, colors, and layout. Get feedback from two classmates. Write one thing you would improve if you had more time.

Chapter 2 Review

Learning Objectives

- Review all graphics concepts from Chapter 2
- Demonstrate integrated skills in a final project
- Reflect on learning and identify growth areas

Engage (5-7 minutes)

Before the harvest is complete, farmers check every crop, celebrate the good ones, and plan for the next season. Today we do the same with our graphics journey—review our skills, celebrate our creations, and plan what to learn next.

Explore (10-12 minutes)

Work in pairs to create a review poster. Each pair chooses five topics from Chapter 2 and creates a visual summary with examples. Draw or demonstrate each concept: raster vs vector, tools, colors, symmetry, layers, effects, text, and formats. Present your poster to another pair.

Explain (10-12 minutes)

Chapter 2 taught us the complete graphics creation process. We started with understanding graphic types, learned tools, colors, symmetry, layers, effects, text, and file formats. Each skill builds on the previous ones to create complete digital art.

Elaborate (8-10 minutes)

Create a final portfolio piece combining everything: use multiple layers, symmetric design, text, effects, and save in the best format. Write an artist's statement explaining your creative process. Display all class projects and have a virtual gallery walk to appreciate each other's work.

Evaluate (5-8 minutes)

Exit ticket: List three skills you mastered in Chapter 2 and one you want to improve. Write one real-life application of graphics skills in your village. Rate your overall confidence from 1 to 5 and explain your rating.

Types of Communication

Learning Objectives

- Define communication and its purpose
- Identify verbal, non-verbal, and digital communication
- Compare traditional and modern communication methods

Engage (5-7 minutes)

Think of how news travels in a village—someone shouts at the market, elders signal with hand gestures, or a message is sent on WhatsApp. All these are forms of communication. Today we explore the different ways people share information.

Explore (10-12 minutes)

Work in pairs. List three ways your family communicates daily—speaking, writing letters, calling on phone, sending messages. Classify each as verbal, non-verbal, or digital. Discuss which method is fastest and why.

Explain (10-12 minutes)

Communication is the exchange of information between sender and receiver. Verbal communication uses spoken words, non-verbal uses gestures and signs, and digital uses technology like phones and computers. Each type has its own advantages depending on the situation.

Elaborate (8-10 minutes)

Imagine your grandfather needs to inform relatives about a wedding. He could send verbal messages through neighbours, write invitation letters, or post on social media. Write which combination of methods he might use and explain your reasoning.

Evaluate (5-8 minutes)

Exit ticket: Name one example each of verbal, non-verbal, and digital communication. Write one advantage of digital communication over traditional methods.

Audio Communication

Learning Objectives

- Understand how sound is used for communication
- Identify devices used for audio communication
- Explain the difference between analog and digital audio

Engage (5-7 minutes)

Think of the temple bell ringing across the valley—its sound carries a message to everyone within earshot. Audio communication uses sound to convey information. Today we learn how audio travels and the technology behind it.

Explore (10-12 minutes)

Listen to different sounds in the school—a whistle, a song, a voice message. Note how each carries a different meaning. Draw a simple diagram showing how sound travels from a speaker to your ear through air.

Explain (10-12 minutes)

Audio communication uses sound waves to transmit information. Microphones convert sound into electrical signals, and speakers convert them back. Analog audio is continuous like flowing water, while digital audio is measured in steps like stairs.

Elaborate (8-10 minutes)

Your teacher records instructions on a phone and sends them to the class group. Trace the journey of that audio message—from voice to microphone, through digital processing, to your phone speaker. Write each step in order.

Evaluate (5-8 minutes)

Exit ticket: Name two devices used for audio communication. Explain in one sentence how digital audio differs from analog audio.

Video Communication

Learning Objectives

- Understand how video combines audio and visual elements
- Identify devices and platforms for video communication
- Explain the basic components of a video file

Engage (5-7 minutes)

Think of watching a folk dance performance on your phone versus seeing it live. Video brings both sight and sound together, making distant events feel real. Today we explore how video communication works.

Explore (10-12 minutes)

Open a video on a phone or computer. Pause it and identify what you see—moving images, text, audio. Count how many frames appear per second by tapping slowly. Discuss with a partner what makes video different from a photo.

Explain (10-12 minutes)

Video communication combines moving images with audio to create rich content. Cameras capture frames—still images shown rapidly to create motion. Platforms like YouTube, video calls, and TV all use video to communicate across distances.

Elaborate (8-10 minutes)

Your school wants to create a video message for Children's Day. List what equipment you need—camera, microphone, lighting—and plan the steps: scripting, recording, editing, and sharing. Draw a simple storyboard with four scenes.

Evaluate (5-8 minutes)

Exit ticket: Name two devices used for video communication. List one advantage of video over audio-only communication.

Introduction to Audacity

Learning Objectives

- Identify Audacity as a free audio editing tool
- Navigate the Audacity interface and key controls
- Record and play back a short audio clip

Engage (5-7 minutes)

Think of a radio jockey who trims songs, adds effects, and mixes voices before broadcasting. Audacity lets you do the same on a computer—for free. Today we meet this powerful audio editing tool.

Explore (10-12 minutes)

Open Audacity on the computer. Explore the toolbar—find the record button, play button, pause, and stop. Click around (but don't save anything) to see what each button does. Write the name and purpose of three buttons you discover.

Explain (10-12 minutes)

Audacity is free, open-source software for recording and editing audio. Its interface has a timeline where audio waves appear, transport controls for play/record/stop, and toolbars for effects and editing. Think of it as a digital tape recorder you can cut and modify.

Elaborate (8-10 minutes)

Record yourself saying a five-word sentence about your village. Play it back and observe the waveform that appears. Identify which parts are loud and which are quiet by looking at the wave heights. Write your observations.

Evaluate (5-8 minutes)

Exit ticket: Name three buttons in Audacity and their functions. Explain what a waveform represents in Audacity.

Recording Audio

Learning Objectives

- Set up microphone and adjust recording levels
- Record clear audio with proper techniques
- Identify and reduce common recording errors

Engage (5-7 minutes)

Think of a radio announcer speaking clearly into a microphone in a quiet room. Good recording requires the right setup—quiet space, proper mic distance, and correct levels. Today we learn to capture clean audio.

Explore (10-12 minutes)

Test recording with a partner. Speak at different distances from the microphone—too close, too far, just right. Record each and listen back. Note which sounds clearest and which has distortion or echo.

Explain (10-12 minutes)

Good recording starts with a quiet environment—close windows, turn off fans. Position the microphone 15-20 cm from your mouth. Adjust input levels so the meter stays in the green zone, never hitting red. Test with a short recording before starting the real one.

Elaborate (8-10 minutes)

Record a 30-second introduction about yourself. Listen back and identify one problem—background noise, too soft, or too loud. Re-record applying the correction. Compare both versions and write what improved.

Evaluate (5-8 minutes)

Exit ticket: List three things to check before recording audio. Explain why recording levels should stay in the green zone.

Editing Audio

Learning Objectives

- Cut, copy, and paste audio segments in Audacity
- Remove unwanted sections and silence from recordings
- Apply basic trimming to clean up audio

Engage (5-7 minutes)

Think of a radio play where different voices, sounds, and music are joined together smoothly. The editor cuts out mistakes and joins the best parts. Today we learn to edit audio like a radio producer.

Explore (10-12 minutes)

Open a recorded audio file in Audacity. Use the selection tool to highlight a section, then cut it. Paste it somewhere else. Try the silence tool on a noisy part. Describe what happens after each action.

Explain (10-12 minutes)

Editing audio is like editing text—you select unwanted parts and remove them. The cut tool removes a section, paste places it elsewhere, and silence replaces sound with quiet. Use undo (Ctrl+Z) if you make a mistake. Always keep the original file as backup.

Elaborate (8-10 minutes)

Take a recording with three mistakes—stutter, long pause, and background noise. Use editing tools to fix each: cut the stutter, trim the pause, silence the noise. Save the cleaned version and compare the before and after.

Evaluate (5-8 minutes)

Exit ticket: Name three editing tools in Audacity and what each does. Explain why you should keep the original recording before editing.

Effects and Filters

Learning Objectives

- Apply audio effects like echo, reverb, and amplification
- Use noise reduction to improve audio quality
- Understand when and how to use different filters

Engage (5-7 minutes)

Think of the echo when you shout in a temple corridor versus the flat sound in an open field. Effects change how audio sounds—adding depth, clarity, or drama. Today we explore Audacity's sound effects.

Explore (10-12 minutes)

Record a short clip and apply three different effects—echo, amplify, and change pitch. Listen to each result. Write which effect made the audio louder, which added repetition, and which changed the voice tone.

Explain (10-12 minutes)

Effects modify audio characteristics. Amplify makes sounds louder or quieter. Echo adds repeating reflections. Reverb creates a spacious feeling. Noise reduction removes background hiss. Apply effects gently—too much can make audio sound unnatural.

Elaborate (8-10 minutes)

Your recording has background fan noise. Use noise reduction to clean it up—first select a noise-only section, get noise profile, then apply to the whole clip. Then add slight reverb to make the voice sound fuller. Save both versions to compare.

Evaluate (5-8 minutes)

Exit ticket: Name three audio effects and their purposes. Explain one situation where noise reduction is essential.

Mixing Audio

Learning Objectives

- Combine multiple audio tracks into one
- Adjust volume levels and balance between tracks
- Use the timeline to align audio elements

Engage (5-7 minutes)

Think of a film song where voices, drums, flutes, and background music blend perfectly. A music mixer balances all these tracks so nothing overpowers the other. Today we learn to mix audio like a studio engineer.

Explore (10-12 minutes)

Open two audio files in Audacity—one voice recording and one background music. Select both and play them together. Notice which is louder. Use the volume slider on each track to balance them. Write what settings worked best.

Explain (10-12 minutes)

Mixing combines multiple tracks into a single output. Each track has independent volume and pan controls. The timeline shows all tracks stacked vertically—align them so sounds happen at the right time. Balance means no single track dominates the mix.

Elaborate (8-10 minutes)

Create a podcast-style clip: your voice narrating over soft background music. Import both tracks, lower the music volume to 30%, keep voice at 100%. Align the start points. Mix and export the final result.

Evaluate (5-8 minutes)

Exit ticket: Explain what mixing means in audio editing. Name two controls used to balance tracks in a mix.

Exporting Audio

Learning Objectives

- Export edited audio in common formats like MP3 and WAV
- Choose appropriate quality settings for different uses
- Understand file size versus quality trade-offs

Engage (5-7 minutes)

Think of saving grain after harvest—you store it in different containers depending on use: large bags for market, small packets for daily use. Audio export is similar—choose the right format and quality for your purpose.

Explore (10-12 minutes)

Export the same recording in two formats—WAV and MP3. Check the file sizes. Play both and listen for differences. Write which file is larger, which sounds better, and which you would send on WhatsApp.

Explain (10-12 minutes)

WAV is uncompressed—full quality but large files, like carrying grain in open sacks. MP3 is compressed—smaller files with slight quality loss, like packing grain in sealed packets. For web sharing, use MP3 at 128-192 kbps. For archiving, use WAV.

Elaborate (8-10 minutes)

Your school needs two versions of the morning announcement: one high-quality for the PA system (WAV) and one small for the school website (MP3). Export both with appropriate settings. Record the file sizes and note the difference.

Evaluate (5-8 minutes)

Exit ticket: Name two audio export formats and when to use each. Explain what happens to file size when you increase audio quality.

Video Editing Basics

Learning Objectives

- Import video clips into an editing application
- Trim and arrange clips on a timeline
- Add transitions and text overlays to video

Engage (5-7 minutes)

Think of a wedding video where photos, clips, and music flow together with smooth transitions. The editor arranges scenes, trims extra footage, and adds titles. Today we learn the basics of video editing.

Explore (10-12 minutes)

Import two short video clips into a simple editor. Trim the beginning and end of each. Drag them onto the timeline side by side. Add a simple transition between them. Write the steps you followed.

Explain (10-12 minutes)

Video editing arranges clips in sequence on a timeline. Trim removes unwanted parts from start or end. Transitions like fade or dissolve create smooth changes between clips. Text overlays add titles or captions. Think of it as arranging photos in an album, but with moving pictures.

Elaborate (8-10 minutes)

Create a 30-second slideshow of your school using five photos. Add text labels for each location, a simple transition between slides, and background music. Export the final video and share with a classmate.

Evaluate (5-8 minutes)

Exit ticket: Name two video editing operations and their purposes. Explain the difference between trimming and splitting a clip.

Project - Audio Story

Learning Objectives

- Plan and script a short audio story
- Record, edit, and mix voice narration with sound effects
- Export and present the final audio story

Engage (5-7 minutes)

Think of a grandmother telling bedtime stories—her voice, pauses, and dramatization create magic. Today you become that storyteller using technology. Create a 2-minute audio story combining voice, music, and effects.

Explore (10-12 minutes)

Brainstorm story ideas with your group. Write a simple script with narration, dialogue, and sound effect cues. Assign roles—narrator, character voices, sound manager. Plan which effects to record or find.

Explain (10-12 minutes)

A good audio story needs a clear script, varied voices, and 恰当 sound effects. Record narration first as the backbone. Add character voices on separate tracks. Layer sound effects and music last. Mix everything so voices are clear above background sounds.

Elaborate (8-10 minutes)

Record your group's story in three stages: narration, dialogue, and effects. Edit each track—remove mistakes, add pauses for drama. Mix them together with balanced volumes. Export as MP3 and present to the class.

Evaluate (5-8 minutes)

Peer evaluation: Listen to another group's story. Rate clarity, creativity, and sound quality on a scale of 1-5. Write one strength and one suggestion for improvement.

Chapter 3 Review

Learning Objectives

- Review all concepts from Audio & Visual Communication
- Demonstrate audio recording, editing, and mixing skills
- Apply chapter knowledge to a practical scenario

Engage (5-7 minutes)

Think of a village fair where different stalls showcase different skills—pottery, weaving, cooking. Today we visit each stall of Chapter 3 and show what we have learned about audio and video communication.

Explore (10-12 minutes)

Complete a quiz matching communication types to examples, Audacity tools to their functions, and audio formats to their uses. Work in pairs and discuss any questions you find difficult.

Explain (10-12 minutes)

Chapter 3 covered types of communication, audio and video technology, Audacity for recording and editing, effects and mixing, exporting, and video basics. Each skill builds on the previous—recording leads to editing, editing to mixing, mixing to export.

Elaborate (8-10 minutes)

Scenario: Your school needs a 2-minute announcement about the annual day. Plan the entire workflow—script writing, recording, editing, adding background music, and exporting. Write each step and the tools you would use.

Evaluate (5-8 minutes)

Self-assessment: Rate your confidence in each skill—recording, editing, effects, mixing, exporting—on a scale of 1-5. Identify one skill to improve and write how you will practice it.

Introduction to Data

Learning Objectives

- Define data and distinguish between data and information
- Identify different types of data in daily life
- Explain why computers represent data as numbers

Engage (5-7 minutes)

Think of a farmer counting sacks of grain—each number is raw data. When he adds them up and says “I have 50 sacks worth ₹25,000,” that becomes useful information. Today we learn what data is and how computers handle it.

Explore (10-12 minutes)

Look around the classroom and list 10 pieces of raw data—names, roll numbers, heights, dates, temperatures. Sort them into categories: numbers, text, and dates. Discuss with a partner which ones become information when combined.

Explain (10-12 minutes)

Data is raw, unprocessed facts—numbers, letters, or symbols. Information is processed data that has meaning. Computers store everything as numbers because their circuits understand only on and off states. Text, images, and sounds are all converted to numbers internally.

Elaborate (8-10 minutes)

Your school attendance register contains data—names, dates, present/absent marks. Write how this raw data becomes information: “Class 6 had 95% attendance this month.” Identify three more examples where data becomes information in daily life.

Evaluate (5-8 minutes)

Exit ticket: Define data and information with one example each. Explain in one sentence why computers use numbers to represent all types of data.

Number Systems

Learning Objectives

- Understand the decimal (base-10) number system
- Introduction to binary (base-2) number system
- Compare how different number systems represent values

Engage (5-7 minutes)

Think of counting on fingers versus using an abacus—both represent the same numbers differently. Computers use binary like a abacus with only two positions: on and off. Today we explore how numbers work in different systems.

Explore (10-12 minutes)

Write the numbers 1 to 10 in normal decimal. Then try to write them using only 0s and 1s—binary. Work with a partner to find patterns. How many binary digits do you need to represent the number 5?

Explain (10-12 minutes)

Decimal uses ten digits (0-9) and is base-10 because we have ten fingers. Binary uses only two digits (0 and 1) and is base-2 because computers have two states—on and off. Each position in binary represents a power of 2, just as each decimal position represents a power of 10.

Elaborate (8-10 minutes)

Convert your roll number to binary. For example, roll number 13 becomes 1101 in binary because $8+4+0+1=13$. Write a table showing decimal 1-8 and their binary equivalents. Look for the pattern in how binary counts.

Evaluate (5-8 minutes)

Exit ticket: Convert decimal 7 to binary. Explain why computers use binary instead of decimal.

Binary Conversion

Learning Objectives

- Convert decimal numbers to binary
- Convert binary numbers to decimal
- Verify conversions using the doubling method

Engage (5-7 minutes)

Think of measuring land in acres versus hectares—same area, different units. Binary and decimal are like that for numbers. Today we learn to convert between them like translating between two languages.

Explore (10-12 minutes)

Use the division method: divide 25 by 2 repeatedly and write remainders. Read remainders bottom-up to get binary. Then use the addition method: convert 1101 back to decimal by adding $8+4+0+1$. Verify both give the same answer.

Explain (10-12 minutes)

To convert decimal to binary: divide by 2, record remainder, repeat until quotient is 0. Read remainders from bottom to top. To convert binary to decimal: multiply each digit by its position value (1, 2, 4, 8, 16...) and add results. Always verify by converting back.

Elaborate (8-10 minutes)

Convert these numbers: decimal 30 to binary, binary 10110 to decimal. Then try a challenge—find the binary for your age. Use the verification method to check each answer is correct.

Evaluate (5-8 minutes)

Exit ticket: Convert decimal 19 to binary and binary 1100 to decimal. Show one conversion step-by-step.

Text Representation

Learning Objectives

- Understand ASCII encoding for English text
- Explain how Unicode supports multiple languages
- Convert text to binary using character codes

Engage (5-7 minutes)

Think of a code language you use with friends—each letter 替换为 a number only you understand. Computers do the same with text—each letter becomes a number. Today we decode how computers read words.

Explore (10-12 minutes)

Look up ASCII values for your name—A=65, B=66, a=97, b=98. Write the binary for each letter in your name. Count how many bits are needed for one character. Discuss why computers need codes for letters.

Explain (10-12 minutes)

ASCII maps 128 characters to numbers—uppercase A-Z (65-90), lowercase a-z (97-122), digits 0-9 (48-57). Unicode extends this to include Hindi, Kannada, and every world script. Each character has a unique code point that computers store as binary.

Elaborate (8-10 minutes)

Write your name in ASCII decimal values, then convert each to binary. For example, “RAVI” becomes 82 65 86 73 in decimal. Discuss why English text needs 7-8 bits per character while Kannada might need more.

Evaluate (5-8 minutes)

Exit ticket: What does ASCII stand for? Write the ASCII decimal value for letter ‘K’ and its binary equivalent.

Image Representation

Learning Objectives

- Understand how images are made of pixels
- Explain color depth and resolution
- Calculate storage needed for a simple image

Engage (5-7 minutes)

Think of a rangoli pattern made of tiny colored dots—step back and you see the full picture. Digital images work the same way—thousands of colored pixels form the complete image. Today we explore how computers store pictures.

Explore (10-12 minutes)

Open a photo on a computer and zoom in until you see individual squares—those are pixels. Count pixels along the width and height. Discuss how more pixels make clearer images. Draw a 4x4 grid and color each cell to make a simple pixel art.

Explain (10-12 minutes)

Images are grids of pixels—each pixel stores a color value. Resolution is pixel count—640x480 means 640 columns and 480 rows. Color depth determines how many colors each pixel can show—8-bit means 256 colors, 24-bit means 16 million colors. More pixels and more colors mean larger files.

Elaborate (8-10 minutes)

Calculate storage for a 100x100 pixel image with 8-bit color: $100 \times 100 \times 8 = 80,000$ bits. Convert to bytes by dividing by 8. Then calculate for 24-bit color. Discuss why high-resolution photos take more space on the phone.

Evaluate (5-8 minutes)

Exit ticket: Define resolution and color depth. Calculate the total pixels in an 800x600 image.

Audio Representation

Learning Objectives

- Understand how sound waves become digital data
- Explain sampling rate and bit depth in audio
- Calculate file size for different audio quality settings

Engage (5-7 minutes)

Think of taking measurements of river water level every hour versus every minute—more frequent measurements give a clearer picture of changes. Audio sampling works the same way—measuring sound wave height many times per second.

Explore (10-12 minutes)

Use Audacity to record a 5-second clip. Check the project rate—usually 44100 Hz. Record another at 8000 Hz. Compare file sizes and sound quality. Write which sounds clearer and which file is smaller.

Explain (10-12 minutes)

Analog sound waves are continuous. Digital audio captures snapshots called samples—thousands per second. Sampling rate is samples per second—44100 Hz means 44,100 measurements each second. Bit depth is sample precision—16-bit means 65,536 possible levels. Higher rate and depth mean better quality but larger files.

Elaborate (8-10 minutes)

Calculate file size for 1 minute of audio: $44100 \text{ samples/sec} \times 16 \text{ bits} \times 60 \text{ sec} = 42,336,000 \text{ bits}$. Convert to bytes, then megabytes. Repeat for 8000 Hz and 8-bit audio. Create a table comparing quality and file sizes.

Evaluate (5-8 minutes)

Exit ticket: Define sampling rate and bit depth. Explain why CD-quality audio uses 44100 Hz sampling rate.

Data Processing

Learning Objectives

- Understand the input-process-output cycle
- Identify different types of data processing
- Explain how computers manipulate data automatically

Engage (5-7 minutes)

Think of a flour mill: farmer brings wheat (input), mill grinds it (process), and flour comes out (output). Computers work the same way with data. Today we explore how computers process information step by step.

Explore (10-12 minutes)

List five daily activities that follow input-process-output: cooking a meal, doing homework, washing clothes. For each, identify the input, the process, and the output. Share your list with the class.

Explain (10-12 minutes)

Data processing follows the IPO cycle: Input collects raw data, Process transforms it using instructions, Output delivers results. Computers perform arithmetic, comparison, sorting, and searching operations on data automatically and at incredible speed.

Elaborate (8-10 minutes)

A shopkeeper enters item prices (input), computer calculates total with tax (process), and prints a bill (output). Design a flowchart for a school marks system: input marks, calculate grades, output report cards. Draw it step by step.

Evaluate (5-8 minutes)

Exit ticket: Define the IPO cycle with one real-world example. Name two types of processing operations a computer can perform.

Data Storage

Learning Objectives

- Identify different storage devices and their uses
- Compare primary and secondary storage
- Understand units of digital storage from bytes to gigabytes

Engage (5-7 minutes)

Think of storing grain: small jars for daily use (RAM), large bins for the season (hard disk), and sealed packets for years (cloud). Computers store data similarly in different places for different purposes.

Explore (10-12 minutes)

List all storage devices you can find or have used: pen drives, memory cards, hard disks, phones. For each, estimate how much it can hold. Create a chart comparing their sizes and capacities.

Explain (10-12 minutes)

Primary storage (RAM) is temporary and fast like a working desk. Secondary storage (hard disk, SSD) is permanent like a cupboard. Units progress: 1 Byte = 8 bits, 1 KB = 1024 Bytes, 1 MB = 1024 KB, 1 GB = 1024 MB, 1 TB = 1024 GB.

Elaborate (8-10 minutes)

Calculate storage needs: a text file is 5 KB, a photo is 3 MB, a song is 5 MB, a video is 500 MB. How many of each can fit on an 8 GB pen drive? Write your calculations and discuss which type uses the most space.

Evaluate (5-8 minutes)

Exit ticket: Name the difference between RAM and hard disk. List storage units from smallest to largest.

Data Compression

Learning Objectives

- Understand why data compression is needed
- Distinguish between lossless and lossy compression
- Identify compressed file formats in daily use

Engage (5-7 minutes)

Think of packing clothes in a suitcase: rolling clothes tightly fits more in the same space. Data compression does the same—shrinks files so they take less space and travel faster. Today we learn how this magic works.

Explore (10-12 minutes)

Save a photo as BMP, then as JPEG. Compare file sizes. Open both and look for visual differences. Do the same with a text file saved as TXT versus ZIP. Write which formats reduce size and whether quality changes.

Explain (10-12 minutes)

Compression reduces file size by removing or encoding redundant data. Lossless compression keeps all original data like ZIP files—perfect reconstruction. Lossy compression removes some data permanently like JPEG—slight quality loss but much smaller files. Choose based on whether perfect quality matters.

Elaborate (8-10 minutes)

A school website needs photos. Should they use BMP (large, perfect quality) or JPEG (small, slight loss)? Discuss trade-offs for different uses: medical images, social media photos, text documents. Write recommendations for each scenario.

Evaluate (5-8 minutes)

Exit ticket: Explain the difference between lossless and lossy compression with one example each. Name two compressed file formats.

Data Encryption

Learning Objectives

- Understand why data needs protection
- Explain the concept of encryption and decryption
- Identify everyday uses of encryption

Engage (5-7 minutes)

Think of sending a secret letter written in a code only your friend can read. Even if someone intercepts it, they cannot understand it. Data encryption works the same way—it scrambles data so only authorized people can read it.

Explore (10-12 minutes)

Try a simple substitution cipher: replace each letter with one three positions ahead (A becomes D, B becomes E). Write a short encrypted message and exchange with a partner. Try to decode each other's messages. Discuss what makes a code strong.

Explain (10-12 minutes)

Encryption converts readable data (plaintext) into scrambled form (ciphertext) using a key. Decryption reverses it with the same or related key. HTTPS on websites, UPI payments, and WhatsApp messages all use encryption to protect data from hackers during transmission.

Elaborate (8-10 minutes)

When you check bank balance on phone, encryption protects your PIN and account number. List five daily activities that use encryption: online shopping, email, messaging, logging into websites, digital payments. Explain what could happen without encryption.

Evaluate (5-8 minutes)

Exit ticket: Define encryption and decryption. Name two everyday applications that use encryption to protect data.

Project - Data Analysis

Learning Objectives

- Collect and organize data from a survey
- Use a spreadsheet to enter and calculate data
- Create a chart to present findings visually

Engage (5-7 minutes)

Think of a village panchayat collecting data on crop yields to plan next season's water distribution. Data analysis helps make informed decisions. Today you become data analysts for your school.

Explore (10-12 minutes)

Design a survey on a topic you care about: favourite subject, travel time to school, or daily screen time. Create five questions, survey 10 classmates, and record responses in a table. Identify patterns in the collected data.

Explain (10-12 minutes)

Data analysis follows these steps: define questions, collect data through surveys, organize in tables, calculate totals and averages, then visualize with charts. Spreadsheets like LibreOffice Calc automate calculations and create professional charts from your data.

Elaborate (8-10 minutes)

Enter your survey data in a spreadsheet. Use formulas to calculate totals, averages, and percentages. Create a bar chart or pie chart showing results. Write three conclusions based on your findings and suggest one action for the school.

Evaluate (5-8 minutes)

Peer presentation: Show your chart and findings to a partner. Explain your data collection method, one calculation, and one conclusion. Give feedback on clarity and accuracy.

Chapter 4 Review

Learning Objectives

- Review all concepts from Data Representation and Processing
- Demonstrate binary conversion and data representation skills
- Apply chapter knowledge to solve a real-world problem

Engage (5-7 minutes)

Think of a harvest festival where farmers display all their crops together. Today we display everything we learned about data—number systems, text, images, audio, processing, storage, compression, and encryption.

Explore (10-12 minutes)

Complete a revision worksheet covering binary conversion, ASCII values, pixel calculations, audio sampling, compression types, and encryption concepts. Work in pairs and help each other with difficult questions.

Explain (10-12 minutes)

Chapter 4 covered data fundamentals, number systems, binary conversion, how text images and audio are represented digitally, data processing with IPO cycle, storage devices, compression techniques, and encryption for security. All concepts connect—binary underlies everything.

Elaborate (8-10 minutes)

Scenario challenge: A farmer wants to store 500 photos, 100 songs, and text records on a device. Calculate total storage needed, recommend a device, suggest compression for photos, and explain how encryption protects his market price data. Write your complete plan.

Evaluate (5-8 minutes)

Self-assessment: Rate your understanding of each topic on a scale of 1-5: binary, text encoding, image representation, audio representation, data processing, storage, compression, encryption. Identify your strongest and weakest areas.

Introduction to Scratch

Learning Objectives

- Understand what Scratch is and its purpose
- Navigate the Scratch interface and identify key areas
- Create a simple project with a sprite and backdrop

Engage (5-7 minutes)

Imagine you have a puppet show at a village mela. You control the puppets, make them move, speak, and dance. Scratch is like a digital puppet theatre on the computer—you tell the characters what to do using colorful blocks instead of strings. Let us explore this fun tool today.

Explore (10-12 minutes)

Open Scratch on the computer. Look at the stage area, the sprite list, and the block palette. Click on different blocks and see what happens when you drag them together. Try making the cat sprite move by attaching a “move 10 steps” block to a “when green flag clicked” block.

Explain (10-12 minutes)

Scratch is a free programming language developed by MIT for young learners. The interface has a stage where your project runs, sprites which are characters, and blocks which are instructions you snap together. Programs in Scratch are called projects and you save them as .sb3 files.

Elaborate (8-10 minutes)

Think of Scratch like arranging blocks to build a house—each block has a specific job. Now create a project where your sprite moves across the stage. Add a backdrop of a village scene and make the sprite say “Namaste” when clicked. Experiment by adding more blocks.

Evaluate (5-8 minutes)

Exit ticket: Name three parts of the Scratch interface and explain what a sprite is. Draw the Scratch screen from memory and label the stage, block palette, and script area.

Motion Blocks

Learning Objectives

- Identify and use basic Motion blocks in Scratch
- Make a sprite move in different directions
- Control sprite movement using steps and turns

Engage (5-7 minutes)

Think about how a bullock cart moves along a village road—forward, backward, turning at corners. Motion blocks in Scratch work the same way, telling your sprite exactly how to travel across the screen. Today we learn to drive our sprites like cart drivers.

Explore (10-12 minutes)

Open your Scratch project. Drag “move 10 steps” and “turn 15 degrees” blocks to the script area. Click them and watch the sprite respond. Try using “go to x: y:” to teleport the sprite to different positions. Experiment with negative numbers to move backward.

Explain (10-12 minutes)

Motion blocks control where and how a sprite moves. “Move 10 steps” pushes the sprite forward in its current direction, while “turn 15 degrees” rotates it. The “glide” block makes smooth movement and “go to” instantly changes position. X is left-right and Y is up-down on the stage.

Elaborate (8-10 minutes)

Create a project where a sprite travels from one end of a village road to another. Use the glide block for smooth movement and turn blocks to navigate corners. Challenge yourself: make the sprite move in a square pattern using four moves and four turns.

Evaluate (5-8 minutes)

Exit ticket: Write the names of three Motion blocks and what each one does. If a sprite faces right and you use “move 20 steps,” which direction does it go? Explain with a diagram.

Looks Blocks

Learning Objectives

- Use Looks blocks to change sprite appearance
- Switch costumes and backdrops programmatically
- Apply size and visibility effects to sprites

Engage (5-7 minutes)

During a yakshagana performance, dancers wear different masks and costumes to change their character. Looks blocks let your sprite do the same—change its costume, size, or even hide completely. Today we give our sprites the power of disguise.

Explore (10-12 minutes)

Click on the Looks category in Scratch. Try “say Hello” and see a speech bubble appear. Use “switch costume” to change how your sprite looks. Add multiple costumes to your sprite and cycle through them. Try “change size by 25” and watch your sprite grow.

Explain (10-12 minutes)

Looks blocks control the visual appearance of sprites. “Say” creates speech or thought bubbles, “switch costume” changes the sprite’s image, and “change size” makes it bigger or smaller. You can also show or hide sprites and switch backdrops to change the scene.

Elaborate (8-10 minutes)

Create a story where your sprite changes costumes to show different emotions—happy, sad, surprised. Use “switch backdrop” to change from a village morning scene to evening. Make a sprite appear, grow larger, say something, and then hide. Think of it like a magic show.

Evaluate (5-8 minutes)

Exit ticket: Name three things Looks blocks can change. If you want your sprite to tell a joke with a speech bubble, which block would you use? Describe a project that uses at least two different Looks blocks.

Sound Blocks

Learning Objectives

- Add sound effects and music to Scratch projects
- Use Sound blocks to play, stop, and control volume
- Record and use custom sounds in projects

Engage (5-7 minutes)

Think of the temple bells and drums during a village festival. Sound makes everything lively and exciting. Scratch has Sound blocks that let your sprites speak, play music, and create sound effects. Let us bring our projects to life with audio.

Explore (10-12 minutes)

Click on the Sound category. Drag “play sound until done” and “start sound” blocks. Choose different sounds from the library like “meow” or “boing.” Try adjusting the volume with “set volume” and play sounds in a loop. Record your own voice using the record feature.

Explain (10-12 minutes)

Sound blocks let you add audio to your projects. “Play sound until done” finishes one sound before the next action, while “start sound” plays it in the background. You can change volume, pitch, and even create drum beats. Sounds make your animations feel more real and engaging.

Elaborate (8-10 minutes)

Create a project where a sprite introduces itself with a recorded voice message. Add background music using a loop block and sound effects when the sprite moves. Think of a village market scene with vendors calling out—use different sounds for different sprites.

Evaluate (5-8 minutes)

Exit ticket: What is the difference between “play sound until done” and “start sound”? Name two sounds available in the Scratch library. How would you make a sprite say its name with a recorded voice?

Events and Control

Learning Objectives

- Use Event blocks to start scripts
- Apply Control blocks like wait and repeat
- Understand the role of the green flag and key presses

Engage (5-7 minutes)

When the school bell rings, students start their activities. When a farmer hears thunder, he takes shelter. Events are like triggers that start actions. Control blocks tell your program what to do and when. Today we learn how to start and manage our Scratch programs.

Explore (10-12 minutes)

Drag the “when green flag clicked” block—this is the most common starter. Try “when space key pressed” and “when this sprite clicked.” Add a “wait 2 seconds” block between actions to see timing. Use “repeat 10” to loop actions. Connect multiple event blocks to one sprite.

Explain (10-12 minutes)

Event blocks start scripts when something happens—clicking the green flag, pressing a key, or clicking a sprite. Control blocks manage the flow: “wait” pauses execution, “repeat” loops actions, and “forever” keeps running until stopped. Together they make your program interactive and organized.

Elaborate (8-10 minutes)

Build a project where pressing number keys makes different sprites appear. Use “wait” blocks to create a countdown before an action. Create a dance where the sprite repeats a sequence of moves forever. Think of it like a kathakali performance with timed sequences.

Evaluate (5-8 minutes)

Exit ticket: What block starts a Scratch project when you click the green flag? What does the “wait 5 seconds” block do? Create a flowchart showing what happens when you press the space key.

Variables

Learning Objectives

- Create and use variables in Scratch
- Understand variables as containers for storing values
- Use variables to track scores and counts

Engage (5-7 minutes)

Think of a donation box at a temple. You put money in, take it out, and the amount changes. Variables in Scratch work like donation boxes—they store numbers or text that can change while your program runs. Today we learn to use these useful containers.

Explore (10-12 minutes)

Click on Variables and then “Make a Variable.” Name it “score” and watch it appear on the stage. Use “set score to 0” and “change score by 1” blocks. Create a project where clicking a sprite increases the score. Display the variable on stage and watch it update live.

Explain (10-12 minutes)

Variables are named containers that store data. You create them, give them a starting value, and change that value as your program runs. “Set” gives a variable a fixed value, while “change” adds to or subtracts from it. Variables help you keep track of scores, counts, and other changing information.

Elaborate (8-10 minutes)

Build a simple quiz game using variables. Create a “score” variable that increases with correct answers and a “lives” variable that decreases with wrong ones. Show both on the stage. Like a village cricket match scoreboard, your variables track the game progress in real time.

Evaluate (5-8 minutes)

Exit ticket: What is a variable? If you set “score” to 10 and then use “change score by 3,” what is the new value? Name one real-life example of something that works like a variable.

Conditionals

Learning Objectives

- Understand if-then-else logic
- Use conditional blocks to make decisions in programs
- Combine conditions with sensing and operators

Engage (5-7 minutes)

A farmer checks the sky every morning. If it is cloudy, he carries an umbrella. If not, he does not. This is conditional thinking—making decisions based on conditions. Scratch conditionals let your sprites make smart choices too. Let us learn this decision-making skill.

Explore (10-12 minutes)

Drag an “if-then” block from Control. Put a sensing block inside like “touching mouse-pointer?” Add actions inside the if block. Try “if-then-else” to do one thing if true and another if false. Test it by making a sprite follow your mouse or hide when touched.

Explain (10-12 minutes)

Conditionals check whether something is true or false and respond accordingly. “If-then” runs actions only when the condition is met, while “if-then-else” offers two paths. You can combine multiple conditions using “and” and “or” operators. This makes your programs respond intelligently to different situations.

Elaborate (8-10 minutes)

Create a game where a sprite catches falling objects. Use “if touching edge” to bounce, “if key pressed” to move, and “if score equals 10” to show a win message. Like a traffic police directing vehicles based on signals, your sprites respond to different conditions.

Evaluate (5-8 minutes)

Exit ticket: Write a conditional statement using “if-then” for a real-life situation. What is the difference between “if-then” and “if-then-else”? Give an example of a condition you could check in Scratch.

Loops

Learning Objectives

- Use repeat and forever loops
- Understand when to use count-controlled vs event-controlled loops
- Combine loops with other blocks for complex actions

Engage (5-7 minutes)

Think of grinding grain on a stone—you repeat the same circular motion over and over. Loops in Scratch do the same thing, repeating actions automatically so you do not have to write them again and again. Today we learn the power of repetition.

Explore (10-12 minutes)

Use “repeat 10” to make a sprite move in a pattern. Try “forever” to keep an action running endlessly. Combine loops with turns to create spirals and circles. Put a loop inside another loop and see what happens. Time how long a forever loop runs.

Explain (10-12 minutes)

Loops repeat blocks of code. “Repeat” runs actions a set number of times, while “forever” keeps running until you stop the program. Loops save time and make code cleaner. Nested loops put one loop inside another for complex repeating patterns, like a farmer planting rows of crops.

Elaborate (8-10 minutes)

Create a dance animation using loops—make the sprite move forward, turn, and repeat to create a pattern. Use “forever” to keep a character bouncing around the screen. Build a project where a loop changes costumes to create an animation, like frames of a traditional scroll painting.

Evaluate (5-8 minutes)

Exit ticket: What is the difference between “repeat” and “forever”? If you use “repeat 4” with “move 50 steps” and “turn 90 degrees,” what shape does the sprite draw? Why are loops useful in programming?

Sensing

Learning Objectives

- Use Sensing blocks to detect interactions
- Detect mouse position, key presses, and sprite collisions
- Create interactive projects using sensing

Engage (5-7 minutes)

Think of a scarecrow in a field that reacts when birds come near. Sensing blocks let your sprites detect what is happening around them—whether something is touching them, the mouse is moving, or a key is pressed. Today we give our sprites senses like real-world objects.

Explore (10-12 minutes)

Try “touching mouse-pointer?” to make a sprite follow your cursor. Use “distance to” to measure how far sprites are apart. Press keys and use “key pressed?” to trigger actions. Ask questions with “ask and wait” to get user input and store answers in a variable.

Explain (10-12 minutes)

Sensing blocks let sprites detect their environment. You can check if sprites touch each other, sense mouse and key input, get user responses with “ask,” and measure distances and time. These blocks make your projects interactive by responding to what players do.

Elaborate (8-10 minutes)

Create a game where a cat sprite catches mice—use “touching” to detect when they meet. Build a quiz where the sprite asks questions and checks answers. Make a drawing tool where the pen follows the mouse using sensing. Like a village well that tells you how much water is there, sensing tells your sprites about their world.

Evaluate (5-8 minutes)

Exit ticket: Name three things Sensing blocks can detect. How would you make a sprite change color when the mouse touches it? What does the “ask” block do and where is the answer stored?

Operators

Learning Objectives

- Use mathematical operators for calculations
- Apply comparison and logical operators
- Combine operators in expressions and conditions

Engage (5-7 minutes)

Think of a shopkeeper calculating bills—adding prices, comparing amounts, checking if customers have enough money. Operators in Scratch do math and comparisons just like that. Today we learn to make our programs calculate and make smart comparisons.

Explore (10-12 minutes)

Use the Operators category. Try “add,” “subtract,” “multiply,” and “divide” with numbers. Create a variable “answer” and use “set answer to $5 + 3$.” Try comparison operators like “greater than” and “less than.” Combine two conditions with “and” and “or” blocks.

Explain (10-12 minutes)

Operators are tools for calculations and comparisons. Arithmetic operators do math (add, subtract, multiply, divide). Comparison operators check relationships (greater than, less than, equals). Logical operators combine conditions (and, or, not). These work inside conditions and variable assignments.

Elaborate (8-10 minutes)

Build a calculator project where clicking number buttons adds values to a total. Create a quiz where the correct answer is calculated using operators. Use comparisons to check if a score is high enough to win. Like a weighing scale that compares two quantities, operators help your programs compare and calculate.

Evaluate (5-8 minutes)

Exit ticket: What does “ $15 / 3$ ” equal in Scratch? Write a condition checking if a variable “age” is greater than 10. How would you use “and” to combine two conditions?

Project - Animation

Learning Objectives

- Apply all Scratch blocks to create an animation
- Plan and design an animation story
- Debug and refine a complete Scratch project

Engage (5-7 minutes)

Think of the animated stories told on village walls during festivals—characters moving, speaking, and performing. Today you become the director of your own digital animation. You will use everything you learned—motion, looks, sound, events, and more—to create a short animated story.

Explore (10-12 minutes)

Plan your animation on paper first. Choose a theme—a village scene, a folk tale, or a daily life story. Sketch your characters and backdrops. List the actions each sprite will perform. Then open Scratch and start building your story scene by scene, testing as you go.

Explain (10-12 minutes)

A good animation needs planning, building, and testing. Plan your story with a beginning, middle, and end. Choose or create sprites and backdrops. Use Motion for movement, Looks for expressions, Sound for audio, Events to start actions, and Control to manage timing. Debug any problems by testing frequently.

Elaborate (8-10 minutes)

Create a 30-second animation telling a simple story. Use at least three different block categories. Add a backdrop change to show different scenes. Include sound effects and speech bubbles. Test your animation, fix any bugs, and then present it to the class. Like a shadow puppet show, your animation tells a visual story.

Evaluate (5-8 minutes)

Peer review: Watch another student's animation and give two positives and one suggestion. Self-reflection: Write three things you learned and one thing you found challenging. Checklist: Does your animation use Motion, Looks, Sound, Events, and Control blocks?

Chapter 5 Review

Learning Objectives

- Review all Scratch programming concepts
- Demonstrate understanding through a quiz
- Connect programming skills to real-world applications

Engage (5-7 minutes)

Imagine you are a village accountant who must review all the accounts before the annual meeting. Today we review everything we learned about Scratch—from basic blocks to full projects. Think of this as a journey from opening the Scratch window to creating your own animations.

Explore (10-12 minutes)

Quick activity stations: Station 1 makes a sprite move with Motion blocks, Station 2 changes costumes with Looks, Station 3 adds sounds, Station 4 uses loops, and Station 5 creates a mini-project combining blocks. Spend two minutes at each station, then share what you created.

Explain (10-12 minutes)

Scratch programming covers Motion for movement, Looks for appearance, Sound for audio, Events to start scripts, Control for timing and loops, Variables for storing data, Sensing for interaction, and Operators for calculations. Together these let you create interactive animations, games, and stories.

Elaborate (8-10 minutes)

Create a revision poster showing all Scratch block categories with examples. Build a “greatest hits” project using at least one block from every category. Connect Scratch to the real world: traffic lights use conditions, calculators use operators, and automated doors use sensing.

Evaluate (5-8 minutes)

Chapter quiz: Answer ten questions covering all Scratch concepts. Include definitions, block identification, and short-answer questions. Bonus: Create a project that demonstrates three concepts and explain your code to a partner.

What is Software?

Learning Objectives

- Define software and distinguish it from hardware
- Identify different types of software
- Explain how software makes computers useful

Engage (5-7 minutes)

Think of a tractor—it is just metal and parts without a driver and instructions. Hardware is the machine, but software is the instructions that make it work. A computer without software is like a bullock cart without a yoke. Today we learn what software is and why it matters.

Explore (10-12 minutes)

Look at the computer screen. The icons you see, the windows that open, the programs you use—all are software. List every program you can find on the computer. Compare hardware (what you can touch) with software (what you cannot touch but runs on the hardware).

Explain (10-12 minutes)

Software is a set of instructions that tells hardware what to do. Hardware is the physical parts you can touch—keyboard, screen, mouse. Without software, hardware cannot perform any task. Software comes in many forms: operating systems, applications, and utility programs.

Elaborate (8-10 minutes)

Think of your school. The building is hardware, but the teachers and their lessons are the software that makes learning happen. Write five examples of software you use daily and explain what each one does. Discuss with a partner why a computer cannot work without software.

Evaluate (5-8 minutes)

Exit ticket: Define software in your own words. Give two examples of hardware and two of software. If hardware is the body, what is software? Explain with a rural Karnataka analogy.

System Software

Learning Objectives

- Identify system software and its types
- Understand the role of the operating system
- Explain how system software manages computer resources

Engage (5-7 minutes)

Think of a village panchayat—it does not grow crops or build houses, but it manages everything so others can do their jobs. System software works the same way—it manages the computer so applications can run. Today we learn about these invisible managers.

Explore (10-12 minutes)

Look at the computer desktop. The operating system is what you see—the taskbar, desktop icons, and start menu. Note down what happens when you turn on the computer: what loads first, what programs start automatically. Identify the operating system name on your computer.

Explain (10-12 minutes)

System software includes the operating system, device drivers, and utility programs. The operating system manages hardware, runs programs, and provides the interface you interact with. Device drivers help the OS communicate with hardware. Examples include Windows, Linux, and Android.

Elaborate (8-10 minutes)

Compare the operating system to a village head who 分配 tasks and resources. The OS allocates memory to programs, manages storage, and handles input-output. Write a paragraph explaining what would happen if your computer had no operating system. How would you use it?

Evaluate (5-8 minutes)

Exit ticket: Name the operating system on your computer. What are two jobs of an operating system? Give one example of system software other than the OS. How is system software different from application software?

Application Software

Learning Objectives

- Identify application software and its categories
- Understand purpose-specific software tools
- Compare application software with system software

Engage (5-7 minutes)

A farmer uses different tools for different jobs—plough for fields, sickle for harvesting, axe for wood. Application software is like these tools—each one is designed for a specific job. Today we explore the many tools available on computers for different tasks.

Explore (10-12 minutes)

Open different programs on the computer: a word processor, a calculator, a web browser. For each, note what task it helps you do. List applications you use at home or school. Categorize them by purpose—writing, browsing, calculating, drawing.

Explain (10-12 minutes)

Application software is designed for specific tasks. Word processors help write documents, spreadsheets handle numbers, browsers access the internet, and media players play videos. Unlike system software which manages the computer, application software helps users do their work. Common examples include MS Word, Chrome, and Paint.

Elaborate (8-10 minutes)

Think of your school subjects—each has a different textbook. Application software is like textbooks for computers. Create a chart matching tasks to software: writing a letter uses a word processor, making a table uses a spreadsheet, drawing uses paint software. Add your own examples.

Evaluate (5-8 minutes)

Exit ticket: Name three types of application software and their purposes. What is the difference between application software and system software? If you want to create a poster, which application would you use? Why?

Word Processing

Learning Objectives

- Open, create, and save documents in a word processor
- Format text with font, size, color, and alignment
- Use basic word processing features for writing

Engage (5-7 minutes)

Think of a scribe writing letters for villagers on nice paper with beautiful handwriting. A word processor is your digital scribe—it helps you write, format, and print documents neatly. Today we learn to use this essential tool for writing.

Explore (10-12 minutes)

Open the word processor. Type a short paragraph about your village. Select text and try changing the font, size, and color. Use bold, italic, and underline. Align text left, center, and right. Save your document with a name you can find later.

Explain (10-12 minutes)

Word processors create and edit text documents. You can type, edit, format, and print text easily. Formatting options include fonts (typeface), sizes, colors, and styles like bold and italic. Alignment positions text on the page. Always save your work regularly to avoid losing it.

Elaborate (8-10 minutes)

Write a one-page letter to a friend describing your school. Use proper formatting: your address at the top, date, greeting, body paragraphs, and closing. Apply different fonts for the heading and body. Center the title and justify the body text. Print a copy if a printer is available.

Evaluate (5-8 minutes)

Exit ticket: How do you save a document? Name two formatting options and explain what they do. If you want to make text larger and red, which formatting tools would you use? Create a formatted heading for a story.

Spreadsheet Basics

Learning Objectives

- Understand the structure of rows, columns, and cells
- Enter and format data in a spreadsheet
- Use basic formulas for calculations

Engage (5-7 minutes)

Think of a shopkeeper's ledger—a big book with rows and columns tracking items sold and money earned. A spreadsheet is a digital ledger that can calculate totals automatically. Today we learn this powerful tool for working with numbers.

Explore (10-12 minutes)

Open the spreadsheet. Note the grid of rows (numbered) and columns (lettered). Click on different cells and see their addresses like A1, B2. Enter numbers in several cells. Try the SUM formula to add a column of numbers. Format cells with borders and colors.

Explain (10-12 minutes)

Spreadsheets organize data in a grid of cells. Each cell has an address based on its column letter and row number. You enter text or numbers in cells and use formulas starting with “=” for calculations. SUM adds numbers, AVERAGE finds the mean, and COUNT tallies items. Formulas update automatically when data changes.

Elaborate (8-10 minutes)

Create a monthly expense tracker. List items in column A and prices in column B. Use SUM to calculate the total. Add a column for quantities and calculate cost as price times quantity. Like a farmer tracking crop yields, your spreadsheet organizes and calculates data automatically.

Evaluate (5-8 minutes)

Exit ticket: What is the address of the cell in column C and row 5? Write a formula to add cells A1 through A5. How is a spreadsheet different from a regular table on paper? Name one advantage of using formulas.

Presentation Software

Learning Objectives

- Create slides with text, images, and backgrounds
- Apply transitions and animations to slides
- Design an effective presentation with clear structure

Engage (5-7 minutes)

Think of a teacher using a chart or poster to explain a lesson to the class. Presentation software is like a digital chart maker—it creates slides that you show one by one to explain ideas. Today we learn to create visual presentations.

Explore (10-12 minutes)

Open the presentation software. Create a new presentation with a title slide. Add a title and subtitle. Create new slides and add text, images, and different backgrounds. Try adding transitions between slides. Reorder slides by dragging them.

Explain (10-12 minutes)

Presentation software creates visual slideshows. Each slide contains content like text, images, and shapes. You choose layouts, themes, and backgrounds for consistent design. Transitions control how slides change, and animations make elements appear or move on slides. Presentations are displayed in slideshow mode.

Elaborate (8-10 minutes)

Create a five-slide presentation about your village. Include a title slide, slides about culture, crops, festivals, and a conclusion. Use different backgrounds for each section. Add at least one image per slide. Practice presenting it to a partner with clear narration.

Evaluate (5-8 minutes)

Exit ticket: What are the three main elements of a good presentation slide? How do you add a new slide? Name the difference between transitions and animations. What makes a presentation easy to understand?

Database Introduction

Learning Objectives

- Understand what a database is and its purpose
- Identify fields, records, and tables
- Recognize real-world uses of databases

Engage (5-7 minutes)

Think of a village voter list—it has names, addresses, ages, and ID numbers organized in rows and columns. A database is like a giant digital voter list that can store, search, and sort thousands of records instantly. Today we learn about this information organizing tool.

Explore (10-12 minutes)

Look at a simple database on the computer. See how data is organized in tables with rows and columns. Each column is a field (like “Name” or “Age”) and each row is a record (one person’s information). Try searching for a specific record and sorting the data by different fields.

Explain (10-12 minutes)

A database stores organized data that can be easily accessed and managed. Data is arranged in tables where columns are fields (types of information) and rows are records (individual entries). Databases let you search, sort, filter, and report data quickly. Examples include school records, library catalogs, and hospital patient lists.

Elaborate (8-10 minutes)

Create a simple database of class students with fields for Name, Roll Number, and Marks. Add at least five records. Practice searching for a student, sorting by roll number, and filtering by marks. Think of how your school office uses a database to track student information.

Evaluate (5-8 minutes)

Exit ticket: What is a database? Define field and record with examples. Give two real-world examples of databases used in Karnataka. Why is a database better than paper records for large amounts of information?

Utility Software

Learning Objectives

- Identify common utility software types
- Understand how utilities maintain computer health
- Use basic utilities like file manager and antivirus

Engage (5-7 minutes)

Think of a village carpenter who has tools not for building furniture, but for maintaining his other tools—oil for hinges, a sharpener for chisels. Utility software is like these maintenance tools—it keeps the computer running smoothly. Today we explore these helpful programs.

Explore (10-12 minutes)

Open the file manager and navigate through folders. Try copying, moving, and deleting files. Look for antivirus software and check if it is running. Find the disk cleanup tool and see how much space you can free up. Try the calculator and notepad utilities.

Explain (10-12 minutes)

Utility software maintains and optimizes computers. File managers organize files and folders. Antivirus software protects against harmful programs. Disk cleanup removes unnecessary files to free space. Backup utilities save copies of important data. These tools keep your computer healthy and efficient.

Elaborate (8-10 minutes)

Practice using the file manager to organize your files into folders—create a folder for each subject. Run a disk cleanup to remove temporary files. Check if antivirus is active. Write a guide for younger students on how to keep computer files organized using utility tools.

Evaluate (5-8 minutes)

Exit ticket: Name three types of utility software and explain what each does. Why is antivirus software important? What happens if you never use disk cleanup? How do you organize files using the file manager?

Software Installation

Learning Objectives

- Understand how software is installed on a computer
- Identify sources of software (CD, download, store)
- Follow safe installation practices

Engage (5-7 minutes)

Think of getting a new tool from the market—you bring it home, unwrap it, and put it in the right place. Installing software is similar—you get it from somewhere, follow steps to set it up, and make it ready to use. Today we learn how software gets onto a computer.

Explore (10-12 minutes)

Watch a software installation demonstration. Note the steps: download or insert media, run the installer, follow on-screen instructions, choose where to install, and wait for completion. Look at the start menu or desktop for newly installed programs. Discuss where software comes from.

Explain (10-12 minutes)

Software installation adds new programs to your computer. Sources include official websites (download), CDs or USB drives, and app stores. Installation usually involves running a setup file, accepting terms, choosing a location, and waiting. Only install software from trusted sources to avoid viruses.

Elaborate (8-10 minutes)

Think of installing software like setting up a new shop in the market. You need space (storage), a signboard (icon), and permissions (license). Create a checklist for safe software installation. Discuss with your partner: why should you never install software from unknown websites?

Evaluate (5-8 minutes)

Exit ticket: Name three sources of software. What are the basic steps to install software? Why is it important to download software only from trusted sources? What could happen if you install software from an unknown website?

Software Licenses

Learning Objectives

- Understand what a software license is
- Differentiate between free and paid software
- Recognize the importance of respecting software licenses

Engage (5-7 minutes)

Think of renting a house—you get permission to live there but do not own it. A software license is like a rental agreement—it gives you permission to use software under certain rules. Today we learn about the legal and ethical side of using software.

Explore (10-12 minutes)

Look at the terms and conditions when you open a program. Find the “About” section in a program to see its license type. Compare free software like Scratch with paid software like MS Office. Discuss what “free” really means in terms of software.

Explain (10-12 minutes)

A software license is a legal agreement that tells you how you can use software. Free and open-source software lets anyone use, study, and share it. Proprietary software requires payment and restricts copying. Freeware is free to use but not open-source. Always respect license terms to use software legally.

Elaborate (8-10 minutes)

Create a chart comparing different license types with examples. Discuss why creators charge for software—it takes time and skill to make. Think about open-source projects like Wikipedia and Scratch that are free. Write three rules for being a responsible software user.

Evaluate (5-8 minutes)

Exit ticket: What is a software license? Name the difference between free software and paid software. Why is it wrong to copy paid software without permission? Give an example of free software you use.

Project - Newsletter

Learning Objectives

- Apply word processing skills to create a newsletter
- Use formatting, images, and layout effectively
- Collaborate and present the final newsletter

Engage (5-7 minutes)

Think of the village notice board where announcements, events, and news are posted for everyone to read. A newsletter is like a digital notice board for your school—sharing news, events, and stories. Today you create a class newsletter using your software skills.

Explore (10-12 minutes)

Plan your newsletter content: what stories, events, or information to include. Decide on sections—news, events, jokes, puzzles. Gather content from classmates. Open the word processor and start creating your newsletter with a title, articles, and images.

Explain (10-12 minutes)

A newsletter combines text, images, and layout to share information. Use word processing skills: titles with large fonts, body text in readable sizes, images for visual interest, and columns for newspaper-style layout. Plan content first, then design. Use borders, colors, and backgrounds to make it attractive.

Elaborate (8-10 minutes)

Work in teams of three to create a one-page class newsletter. Assign roles: editor, designer, and content writer. Use word processing formatting—headers, columns, images, borders. Include at least three articles and two images. Print copies or share digitally with other classes.

Evaluate (5-8 minutes)

Peer review: Read another team's newsletter and evaluate content, design, and readability. Self-assessment: What software skills did you use? Did the project meet its goals? Present your newsletter to the class and explain your design choices.

Chapter 6 Review

Learning Objectives

- Review all software concepts from Chapter 6
- Demonstrate understanding through activities
- Connect software knowledge to everyday computer use

Engage (5-7 minutes)

Think of a harvest festival where farmers display all their crops together. Today we display everything we learned about software—from system software to applications, from word processors to databases. This is our knowledge harvest celebration.

Explore (10-12 minutes)

Quick activity: Station 1 identifies software types, Station 2 practices word processing, Station 3 creates a spreadsheet formula, Station 4 designs a presentation slide, and Station 5 organizes files using the file manager. Complete each station in three minutes and move on.

Explain (10-12 minutes)

Chapter 6 covered software as computer instructions, system software managing hardware, application software for specific tasks, word processors for writing, spreadsheets for numbers, presentations for visuals, databases for organizing data, and utilities for maintenance. Software installation and licenses govern how we obtain and use programs.

Elaborate (8-10 minutes)

Create a software reference guide listing every type of software learned with examples and uses. Build a flowchart showing how to choose the right software for a task. Discuss why understanding software matters—it helps you use computers effectively and responsibly.

Evaluate (5-8 minutes)

Chapter quiz: Twenty questions covering all Chapter 6 topics—definitions, matching, short answers, and scenarios. Include questions about software types, uses, installation, and licenses. Bonus: Explain to a younger student the difference between system and application software.